

The `featuretree` IR: an Open Feature-Tree Format, Emitted Natively into Several CAD Systems and Checked by Re-Reading

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<https://github.com/punkfab/featuretree>

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Abstract

A STEP file carries a part’s shape but not how it was built; a CAD system’s native file carries the history but only that vendor can read it. We publish the specification of a small neutral format between the two: a declarative JSON list of named features — sketches with arcs, pad, pocket, fillet, revolve and placed cuts, with draft — whose references to faces and edges are stored as queries resolved against live geometry, never as kernel identifiers. One IR is emitted as a native, editable feature tree in FreeCAD (headless PartDesign), build123d (code), and Onshape (REST, with every reference written as a FeatureScript query that Onshape re-evaluates on regeneration), and as scripts for Autodesk Fusion and SolidWorks. Every emitted part is checked the same way: exported to STEP and compared with a reference build by volume and a Boolean-free point-sampled IoU, with a per-feature volume trace that names the first feature where two backends disagree. We report the measurements, including the parts that fail. On the eleven feature trees recovered from the NIST MBE PMI corpus FreeCAD reproduces the reference on 10 of 11; Onshape reproduces every feature of our coverage parts exactly, but its corpus run was stopped by the free plan’s daily API quota after two parts. Running the backends against each other found six defects, including one in the reference backend and two rules the specification had never stated. Separately, reading NIST’s native SolidWorks files back through two decoders shows why one decoder is not enough: Onshape’s importer matches NIST’s own STEP on 11 of 11 parts, an open-source decoder on 1, four of its failures passing its own check. Neutral feature exchange is not new — it goes back to the macro-parametric work of 2003, and a closely related system appeared this year — so we claim no novelty. We offer an open specification, an MIT-licensed implementation, and measurements that can be reproduced.

1 Introduction

Moving a parametric part between CAD systems loses its design history. STEP, the neutral exchange standard, has carried shape reliably for decades; its parts for construction history (ISO 10303-55/108/111) were specified but, as their own authors reported, not adopted by commercial translators [29]. The alternatives that did ship — neutral modelling commands replayed through each system’s API [26, 34], and commercial bridges such as Proficiency [31] — showed that the approach works, and identified the hard problems: feature mapping, persistent naming and constraint translation [34].

Kind	Parameters and meaning
sketch	circles (c_x, c_y, r), rects (w, h, c_x, c_y), polys (closed loops of $(x, y[, b])$, b = bulge); on plane XY or XZ , or on a face given by the query {"face_of", "side": top bottom}
pad	extrude a sketch: length , symmetric , taper (deg)
pocket	cut a sketch: through or blind length , taper
fillet	radius on the edges selected by a query, e.g. the largest circle on the top
revolve	revolve an XZ profile about Z by angle
polar_pocket	count cylinders tangent to a circle (roller pockets)
prism_cut	cut a profile drawn on the plane (origin , normal , xdir), depth along the normal, taper

Table 1: Feature kinds. Lengths in millimetres, angles in degrees.

The question has come back for a new reason. Parts are now written by programs and by language-model agents, and most of that output is either code for one kernel (CadQuery, FreeCAD Python) or bare geometry [1, 33, 42]. A design that should open as an editable tree in whichever system its user has needs a representation that is small enough for a program to emit, explicit enough for several CAD systems to rebuild natively, and checkable. CADIR, which appeared in August 2026, targets the same need with an executable Python IR and adapters for FreeCAD, SolidWorks and Fusion [20]. **featuretree** was developed independently (public since June 2026) and makes different choices; we describe them and measure the result.

Contributions.

1. **A specification** (§2): a declarative, non-executable JSON feature list with seven feature kinds, query-based references, fixed coordinate and sign conventions, and well-formedness rules with a validator.
2. **Five backends** (§3): build123d (the reference), FreeCAD and Onshape, which we run and measure, and generated scripts for Fusion and SolidWorks, which we have not yet run.
3. **One verification method for every backend** (§4): STEP export, a volume and Boolean-free IoU gate, a per-feature volume trace, and read-back of native files through two independent decoders.
4. **Measurements** (§5): every backend on the eleven recovered NIST trees and on coverage parts; the six defects that cross-backend disagreement exposed; a two-decoder study of NIST’s native SolidWorks files; and a comparison of the designers’ own trees with ours.

2 The IR

A part is {"name": ..., "features": [...]}: an ordered list of named features. Order is construction order; a feature may refer only to earlier features, and only by name. Table 1 lists the kinds. Listing 1 is a complete part.

Listing 1: A complete part: a plate, a pocket in its top face, and a fillet.

```
{
  "name": "plate", "features": [
    {
      "kind": "sketch", "name": "outline", "plane": "XY",
      "polys": [[[-20,-15],[20,-15,0.4142],[20,15],[-20,15]]],
      {
        "kind": "pad", "name": "body", "sketch": "outline", "length": 10,
        {
          "kind": "sketch", "name": "recess_sk", "on": {
            "face_of": "body", "side": "top"},
            "circles": [[0, 0, 6]]},
            {
              "kind": "pocket", "name": "recess", "sketch": "recess_sk", "through": false, "length": 4},
              {
                "kind": "fillet", "name": "round", "radius": 1, "select": {
                  "circles": "top_outer"}}}}}]
}
```

Backend	Driven by	References become	Status
build123d	in-process Python (OCCT)	queries on the live solid	reference
FreeCAD 1.0	headless <code>freecadcmd</code> , <code>PartDesign</code>	<code>FaceN/EdgeN</code> re-resolved per build	measured
Onshape	REST v6 feature JSON	FeatureScript query strings	measured
Fusion	generated script (<code>adsk.fusion</code>)	queries evaluated in the script	not yet run
SolidWorks	generated script (COM, <code>pywin32</code>)	queries evaluated in the script	not yet run

Table 2: Backends. “Measured” means built and gated in this paper.

References are queries. A face-attached sketch names “the top face” and a fillet names “the largest circular edge on top”, and every backend resolves the query against its own live geometry each time the part is built. This is the approach of generic naming [6] and of geometry-based semantic identifiers [40], and of the selectors in `CadQuery`, `build123d` and `Onshape`’s `FeatureScript`; it avoids tracking kernel names through history [17, 24]. The known cost applies: after a large edit a query can resolve to a different entity.

Conventions. Coordinates of a face-attached sketch stay global: a circle at $(0, 0)$ on the top face is above the origin, whatever frame the backend gives that face. `XZ` maps (u, v) to $(u, 0, v)$. A pad grows $+Z$, or $-Z$ from a bottom face. A bulge is the DXF arc factor $\tan(\theta/4)$, its arc’s midpoint on the chord’s left normal, and $|b| < 10^{-4}$ is a straight line. A positive taper shrinks the profile along the extrusion (for a pocket, as it goes deeper). A sketch’s region is its loops under the nesting rule: a loop inside another is a hole, an island in a hole is material again, and separate loops are separate regions.

Well-formedness. `ir.validate(spec)` checks what every backend is entitled to assume: unique names; references to earlier sketches; closed loops (at least three segments, or two if one is an arc) with no zero-length segment and no self-intersection; within one sketch, loop boundaries that never touch or cross; a taper only on a blind pocket; an orthogonal placement frame. Two of these rules (non-touching loops and the bulge threshold) were added after backends disagreed about parts that broke them (§5.2).

3 Backends

Table 2 summarises them. **build123d** is the reference: every other backend is compared with its build of the same IR. **FreeCAD** writes a native `PartDesign` body in which each feature’s label is its IR name, so a dimension edited in the GUI reads back by name. **Onshape** receives native

Sketch, Extrude, Fillet, Revolve and Plane features. Every reference is a FeatureScript query string rather than an entity identifier: a profile is `qContainsPoint(qSketchRegion(id), p)` for an interior point p of each material region, a face-attached sketch plane is the planar Z -normal face farthest along $\pm Z$, and a fillet’s edges are the circular edges containing points computed from the same rule the reference uses. Onshape re-evaluates them on every regeneration, so the tree stays editable. Placed cuts on oblique planes and polar pockets are not yet emitted. **Fusion** and **SolidWorks** cannot be driven from outside the program, so the emitter writes a self-contained script: the IR as data, the reference volume after every feature, and a small runtime that re-authors each feature and resolves each query inside the program, then exports STEP and a report. The shared geometry those runtimes draw from is tested by replaying it with `build123d`, but the scripts have not yet run in either program, so we report no results for them.

4 Verification

A backend’s output is exported to STEP and compared with the reference build of the same IR. It passes (VERIFIED) when volumes agree within 0.5 % and the intersection-over-union is at least 99.5 %. IoU is estimated by classifying uniformly sampled points against both solids [39], never by a Boolean, because OpenCASCADE’s Booleans returned wrong unions on these parts; robustness failures of this kind are well documented [12, 45]. The volume check is the idea behind CAX-IF’s geometric validation properties [4] and Sap and Shapiro’s invariant-property testing [35]; IoU additionally catches misplaced material that leaves volume unchanged.

Each backend also records the volume after every feature. When a part fails, the first feature whose volume leaves the reference is where to look; every defect in §5.2 was found that way.

For native files (`.sldprt`, `.f3d`) a third check applies: the file is decoded to STEP by two independent decoders, and its stored feature history is matched to the IR by name, kind and driving value. Comparing two implementations of the same function is differential testing [25, 44]; it answers the oracle problem [2] that a single decoder’s own consistency check cannot (§5.3).

5 Results

5.1 FreeCAD and Onshape against the reference

The corpus is the eleven feature trees that `featuretree`’s recogniser recovers from the NIST MBE PMI test parts [19], 2 to 144 features each (the recogniser and its accuracy are the subject of a companion paper; here the trees are simply inputs). Table 3 gives FreeCAD’s result on every part and Onshape’s where it ran.

FreeCAD reproduces the reference on 10 of 11 parts. The exception is FTC-06 (build fails); §5.2 traces the cause. Onshape builds VERIFIED on FTC-11. On CTC-01 it builds 53 of 54 features; one placed cut, whose profile edge lies exactly on the part’s outer wall, is rejected by Onshape’s kernel, and the part reaches 99.36 % IoU, just under the gate. The free Onshape plan’s daily API quota (about 1,500 calls) then stopped the run; the other nine parts need roughly 2,500 calls, and we will add them in a revision rather than delay the rest.

Onshape was also run on coverage parts built to exercise every path it supports (Table 4). With the conventions pinned, every feature’s volume matches the reference to three decimals, with one exception: a pocket whose circle crosses a concave arc edge removes 4.175 mm^3 more in

Part	Features	Valid	FreeCAD			Onshape		
			Verdict	IoU %	$ \Delta V $ %	Verdict	IoU %	$ \Delta V $ %
CTC-01	54	yes	VERIFIED	100.00	0.00	53/54 built	99.36	0.58
CTC-02	144	yes	VERIFIED	99.90	0.05	not run		
CTC-03	64	no	VERIFIED	100.00	0.00	not run		
CTC-04	108	yes	VERIFIED	100.00	0.00	not run		
CTC-05	57	yes	VERIFIED	99.78	0.23	not run		
FTC-06	27	no	build fails			not run		
FTC-07	29	yes	VERIFIED	100.00	0.00	not run		
FTC-08	37	yes	VERIFIED	100.00	0.00	not run		
FTC-09	80	yes	VERIFIED	100.00	0.00	not run		
FTC-10	38	yes	VERIFIED	100.00	0.00	not run		
FTC-11	2	yes	VERIFIED	100.00	0.00	VERIFIED	100.00	0.00

Table 3: Each recovered NIST feature tree built by FreeCAD and Onshape and gated against the build123d reference build of the same IR (VERIFIED: $\text{IoU} \geq 99.5\%$ and $|\Delta V| \leq 0.5\%$). “Valid”: passes `ir.validate`. Onshape’s run stopped at the free plan’s daily API quota.

Onshape than the exact answer, computed independently, which build123d matches. We have not explained it.

5.2 What disagreement found

Running backends against each other is a test of each of them and of the specification. The trace located six defects, none of which any single backend reported:

1. **The reference was wrong.** build123d decided that a sketch was on the bottom face by its height being below zero, so a pad from a bottom face at $z = 0$, the common case, grew upward into the part. Onshape disagreed on the first such feature. The rule now uses the sketch’s declared side. No published result changed: all eleven recovered trees give identical reference volumes before and after.
2. **FreeCAD kept one solid.** A PartDesign body is one solid by default, so a sketch of several separate outlines padded only one of them, exactly halving the first feature of CTC-03 and CTC-04. FreeCAD 1.0’s `AllowCompound` restores the specified union; CTC-03 went from 2.5% IoU to VERIFIED.
3. **A rule the specification lacked: loops must not touch.** Two recovered trees contain sketches whose loops share boundary. build123d unions loops one at a time and accepts them; FreeCAD rejects some (FTC-06). `ir.validate` now reports them; the recogniser should not produce them.
4. **A rule the specification lacked: near-zero bulges.** CTC-02 contains arcs of bulge 5×10^{-6} , radius about two kilometres. FreeCAD’s sketch solver fails on them. Below 10^{-4} a bulge now means a straight line in every backend; the reference volume of CTC-02 moves by $-0.000\,009\%$.
5. **Onshape’s draft direction** is the opposite of our first guess; a tapered boss grew where it should shrink. Pinned by matching the reference on tapered pads and pockets in both directions.

Part	Feature	Onshape (mm ³)	Reference (mm ³)	Difference
plate (sample)	body	11,497.345	11,497.345	+0.000
	hole	11,497.345	11,497.345	+0.000
every kind	base	22,028.273	22,028.273	+0.000
	top_pk	21,784.056	21,788.231	-4.175
	hub	24,328.746	24,332.921	-4.175
	side_slot	24,157.241	24,161.417	-4.176
	thru	24,006.445	24,010.621	-4.176
	round	23,980.157	23,984.332	-4.175
taper A	base	12,000.000	12,000.000	+0.000
	up	12,432.802	12,432.802	+0.000
	down	12,668.025	12,668.025	+0.000
	sym	12,724.573	12,724.573	+0.000
taper B	base	12,000.000	12,000.000	+0.000
	pk_bottom	11,826.944	11,826.944	+0.000
	pk_top	11,632.467	11,632.467	+0.000
	side	11,550.769	11,550.769	+0.000

Table 4: Onshape coverage parts: volume after every solid feature against the build123d reference. The -4.175 offset enters at `top_pk` (a pocket crossing a concave arc edge) and is carried through; everything else agrees to three decimals.

6. **Failure behaviour differs.** A cut that removes nothing is an error in Onshape and silently does nothing in build123d; the trace shows both.

5.3 Reading native files back: one decoder is not enough

NIST publishes the same test parts as native SolidWorks 2018 files. We decoded each to STEP with an open-source clean-room decoder, `cadmepg` 0.6.0, and with Onshape’s commercial importer, and compared both with NIST’s own STEP of the part (Table 5). Onshape matches on 11 of 11 (IoU $\geq 99.93\%$); `cadmepg` on 1. Four of `cadmepg`’s wrong parts pass its own model check and convert without complaint; four more yield a shell that is not a solid. Its decoding of the feature history, on the other hand, looked right on the same files. The practical rule we adopted: a native file’s geometry is judged by the commercial importer when available, its history by the open decoder. We reported the geometry failures upstream with a reproduction.

5.4 Designers’ trees against recovered trees

The same native files carry the designers’ own feature trees. Counting only solid-modifying operations, the designers used 1 to 42 per part; the trees our recogniser recovers from the STEP files are 2 to 4 times longer on the CTC parts (CTC-01: 10 against 39), mostly because one designer hole or pattern feature stands for many of our cuts (Table 5). That gap is a measurable target for intent recovery.

6 Related work

Neutral feature exchange. STEP’s construction-history parts (ISO 10303-55/108/111/112) [14, 16, 29]; the macro-parametric approach and its test rallies between commercial systems [15, 26,

Part	Onshape import		cadmpeg 0.6.0		Solid operations	
	IoU %	ΔV %	IoU %	STEP	Designer	Recovered
CTC-01	100.00	-0.000	21.39	invalid	10	39
CTC-02	99.94	+0.009	3.47	invalid	42	89
CTC-03	100.00	-0.164	11.14	invalid	15	33
CTC-04	100.00	-0.001	22.37	no solid	24	93
CTC-05	100.00	-0.000	6.99	no solid	17	40
FTC-06	100.00	+0.000	25.83	invalid	21	18
FTC-07	99.93	+0.009	0.00	no solid	28	21
FTC-08	100.00	+0.008	6.95	invalid	30	32
FTC-09	100.00	-0.000	80.72	invalid	33	40
FTC-10	100.00	+0.001	26.46	no solid	19	25
FTC-11	100.00	-0.000	100.00	valid	1	1

Table 5: NIST’s native SolidWorks 2018 files decoded to STEP by two decoders and compared with NIST’s own STEP of each part; and the number of solid operations in the designer’s tree (as decoded) against the tree `featuretree` recovers from the STEP.

34]; Proficiency’s universal product representation [31]; BYU’s multi-user neutral database [38]. Interoperability’s cost motivated all of it [5]. **Closest system:** CADIR [20] records a construction graph from an executable Python IR and rebuilds native histories in FreeCAD, SolidWorks and Fusion, matching topology by geometric signature; `featuretree` differs in being a declarative data format, in storing queries rather than matched selections, in targeting Onshape and build123d, and in sharing its IR with a STEP recogniser. **CAD as a language.** Code-CAD (OpenSCAD, CadQuery, build123d, FeatureScript, KCL) [21] and programming-language treatments [7, 27, 28, 32]; learned construction sequences, mostly sketch and extrude [13, 33, 37, 41–43]. **Persistent naming** [3, 6, 8, 17, 24, 40]. **Language models for CAD** emit CadQuery, FreeCAD Python or FeatureScript for one system each [1, 10, 22, 23, 30]. **Exchange quality:** validation properties [4], invariant-property and query-based interoperability [11, 35, 36], benchmarks [9], and NIST’s conformance testing [18, 19].

7 Limitations

- **Fusion and SolidWorks have not run.** Their scripts exist and their geometry is replay-tested, but API behaviour, sign conventions and profile selection are unverified until a run passes the gate.
- **Onshape’s corpus run is incomplete** (2 of 11 parts) because of the free plan’s API quota, and it does not yet emit oblique placed cuts or polar pockets.
- **The reference is OpenCASCADE, and so is FreeCAD.** Their agreement is partly self-consistency. Onshape (Parasolid) is the independent kernel.
- **Single bodies, no sketch constraints, no assemblies** in this IR; sketches arrive in the target unconstrained.
- **Queries can re-resolve wrongly** after large edits; we have not measured how often.
- **The corpus is eleven parts** and the trees are machine-recovered, not hand-authored.

8 Availability

Code, specification (`ir.py`), validator and every backend are MIT-licensed at <https://github.com/punkfab/featuretree>. The measurements are regenerated by:

- `paper/figures/freecad_corpus.py` and `onshape_corpus.py` (Table 3);
- `paper/figures/native_nist.py` (Table 5);
- `paper/ir/tables.py`, which writes every table and inline number.

The software is archived as 10.5281/zenodo.22848721; the companion recovery paper is 10.5281/zenodo.22848792.

How this was built. The work was iterative: we steered the design with Claude (Anthropic) over many rounds, pushing on correctness, generality and performance; Claude wrote and debugged the code and ran the measurements.

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